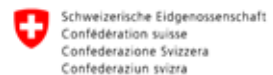


DIGITAL FINANCE

This project has received funding from the Horizon Europe research and innovation programme under the Marie Skłodowska-Curie Grant Agreement No. 101119635



State Secretariat for Education,
Research and Innovation SERI



**Funded by
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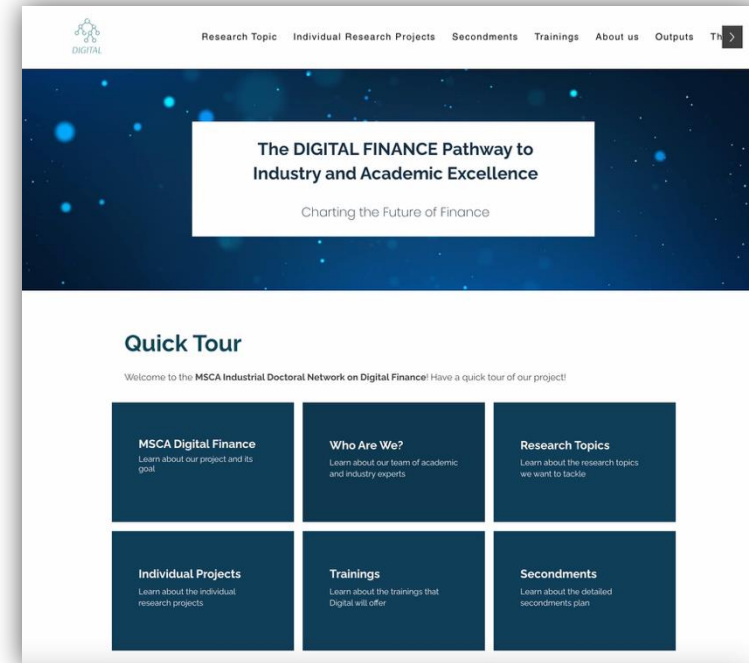
**Research Group: Applied AI
Research & Solutions - AIRS**

Bern Business School

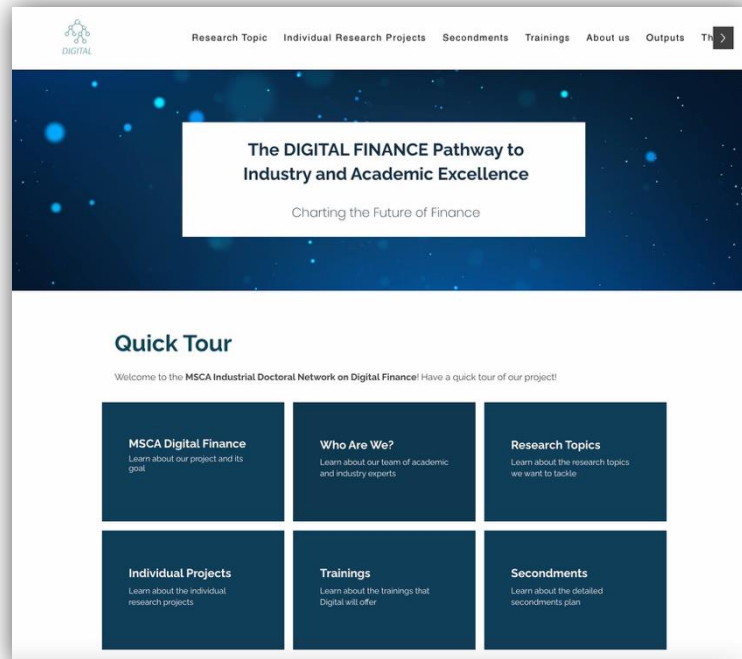




MSCA **Digital Finance**



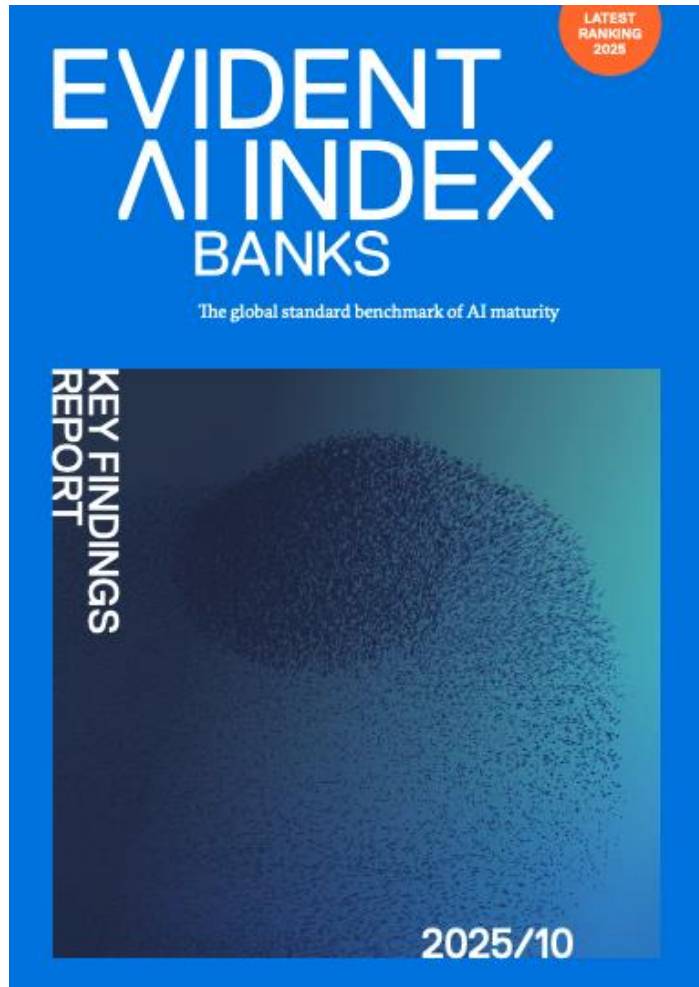
MSCA **Digital Finance**



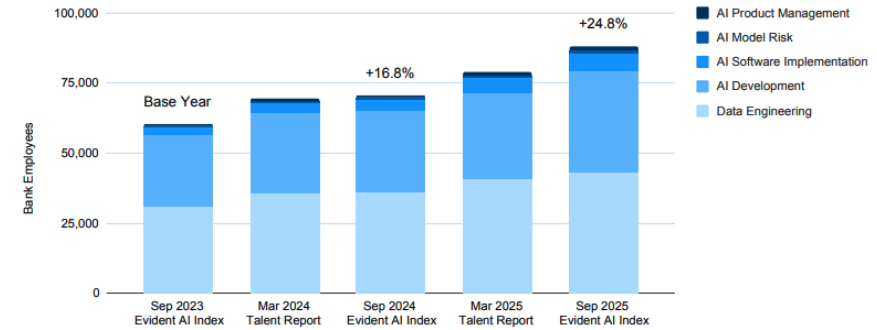
Lead of a Research Work Package –
Towards EXPLAINABLE and FAIR AI-generated decisions

Four Doctoral Candidates

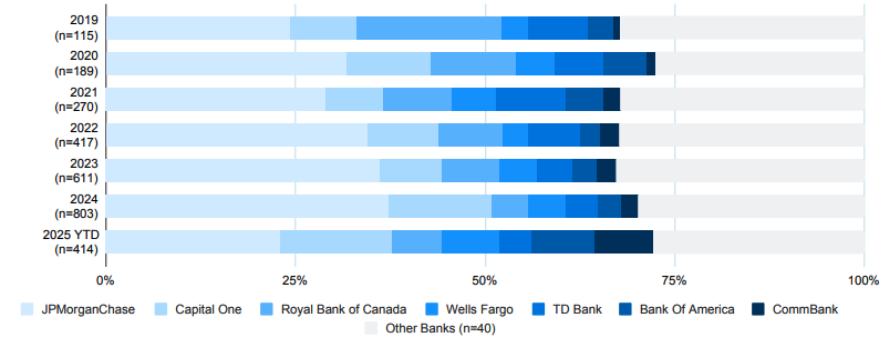
Secondments in ECB, ING, Cardo,
Swedbank, DB ...



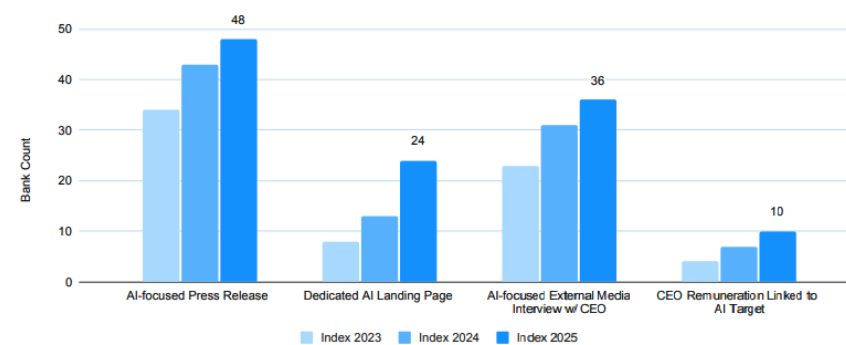
AI TALENT VOLUME, BY CAPABILITY AREA



SHARE OF AI-FOCUSED RESEARCH PAPERS PUBLISHED, BY BANK



SHARE OF BANKS ENGAGING IN SPECIFIED ACTIVITY OR CHANNEL, OVER TIME



General Data Protection Regulation

GDPR

Welcome to gdpr-info.eu. Here you can find the official [PDF](#) of the Regulation (EU) 2016/679 (General Data Protection Regulation) in the current version of the OJ L 119, 04.05.2016; cor. OJ L 127, 23.5.2018 as a neatly arranged website. All Articles of the GDPR are linked with suitable recitals. The European Data Protection Regulation is applicable as of May 25th, 2018 in all member states to harmonize data privacy laws across Europe. If you find the page useful, feel free to support us by sharing the project.

The General Data Protection Regulation (GDPR) mandates that **individuals have the legal right to receive explanations** for decisions made by algorithms that may impact them.

XAI

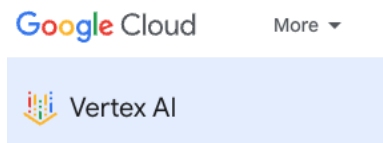
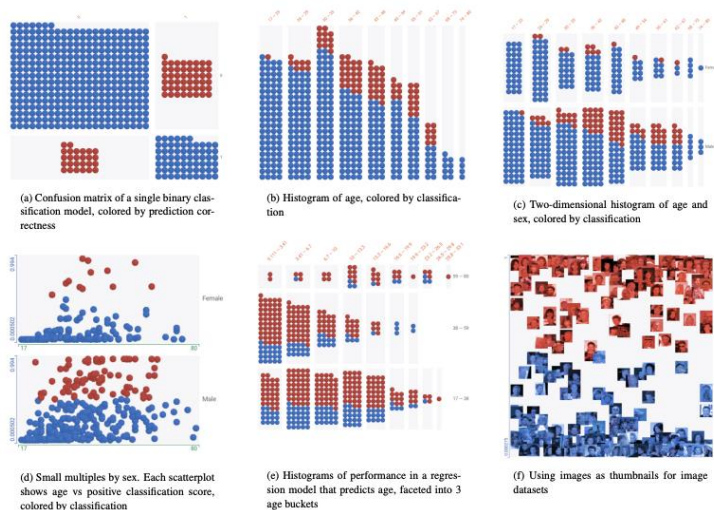
Google Research Who we are Research areas Our work Programs & events Careers Blog

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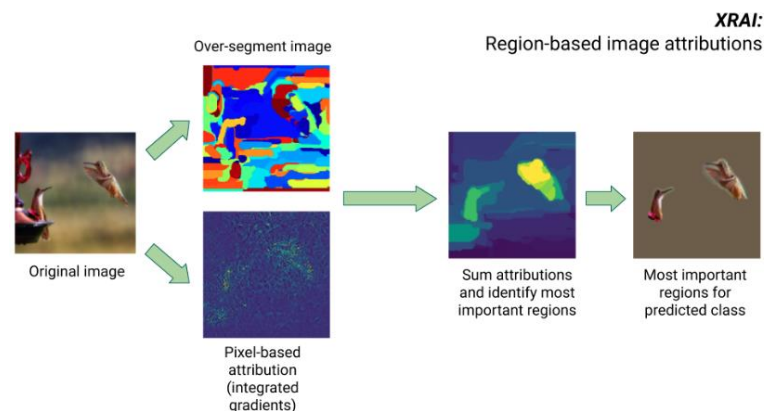
The What-If Tool: Interactive Probing of Machine Learning Models

James Wexler · Mahima Pushkarna · Tolga Bolukbasi · Martin Wattenberg · Fernanda Viégas · Jimbo Wilson · (to appear)

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Introduction to Vertex Explainable AI



InterpretML

Documentation Contribute [GitHub](#)

Understand Models. Build Responsibly.

A toolkit to help understand models and enable responsible machine learning

[Get Started](#) [Learn More](#)



AI Explainability 360 **OLFAI** Quick Start Docs GitHub

AI Explainability 360: Understand how ML models predict labels

The AI Explainability 360 toolkit, an LF AI Foundation incubation project, is an open-source library that supports the interpretability and explainability of datasets and machine learning models. The AI Explainability 360 Python package includes a comprehensive set of algorithms that cover different dimensions of explanations along with proxy explainability metrics. There is no single approach to explainability that works best. The toolkit is designed to translate algorithmic research from the lab into the actual practice of domains as wide-ranging as finance, human capital management, healthcare, and education. IBM moved AI Explainability 360 to LF AI in July 2020.

[Get Started](#) [Join the Conversation](#)

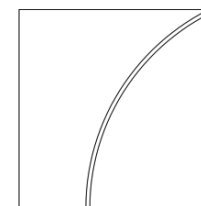
Financial Stability Institute **BIS**

Occasional Paper No 24

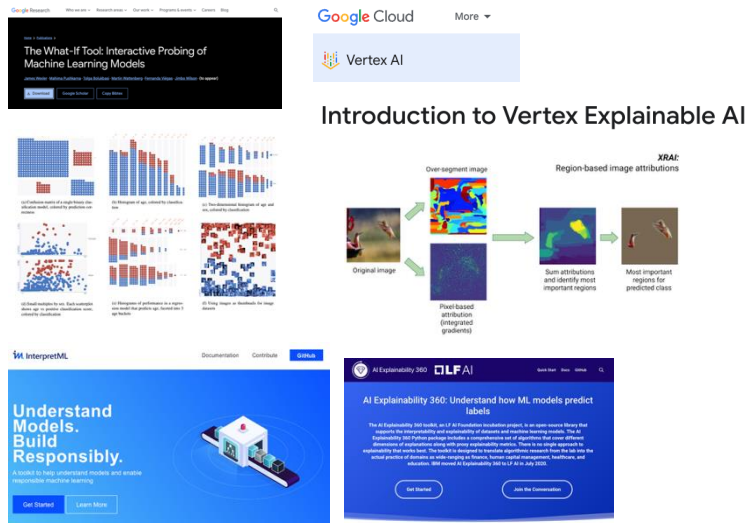
Managing explanations: how regulators can address AI explainability

by Fernando Pérez-Cruz, Jermy Prenio, Fernando Restoy, Jeffery Yong

September 2025



XAI in the MSCA Digital (and BFH)



What **hurdles** remain with respect to explaining AI systems?

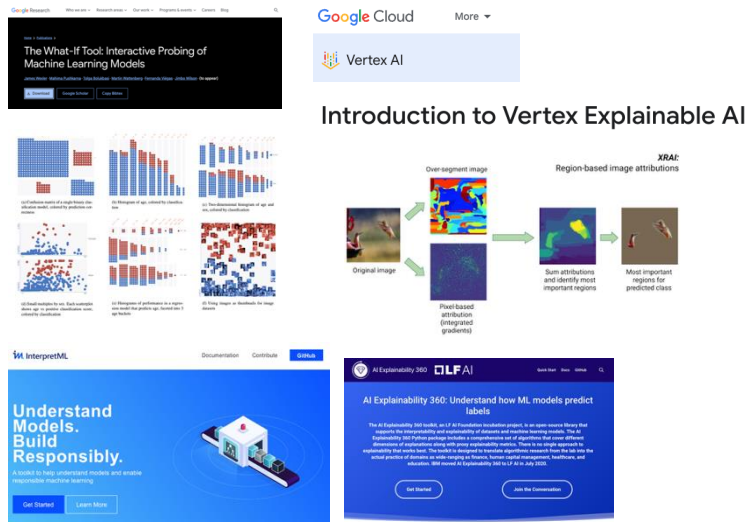
What are **the key limitations of classic XAI methods** when applied to AI-based forecasting systems?

Can we improve the utility of explanations by **grounding model explanations in domain-specific ontologies**?

What are the **explainability requirements of different stakeholders** in finance?

How can we provide more interpretability and transparency in specific use cases in finance – like **portfolio decarbonization**?

XAI in the MSCA Digital (and BFH)



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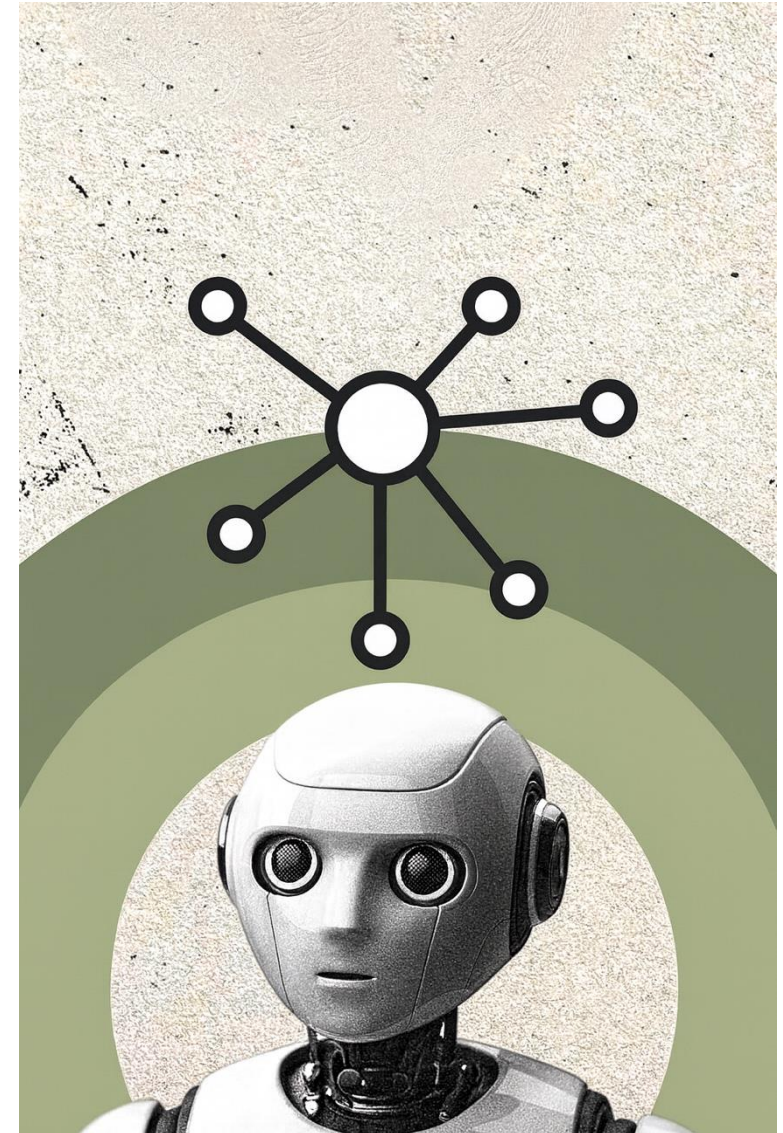
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Rebecca Di Francesco

Investigating the Utility of Classic XAI Methods in Finance

Jens Reil

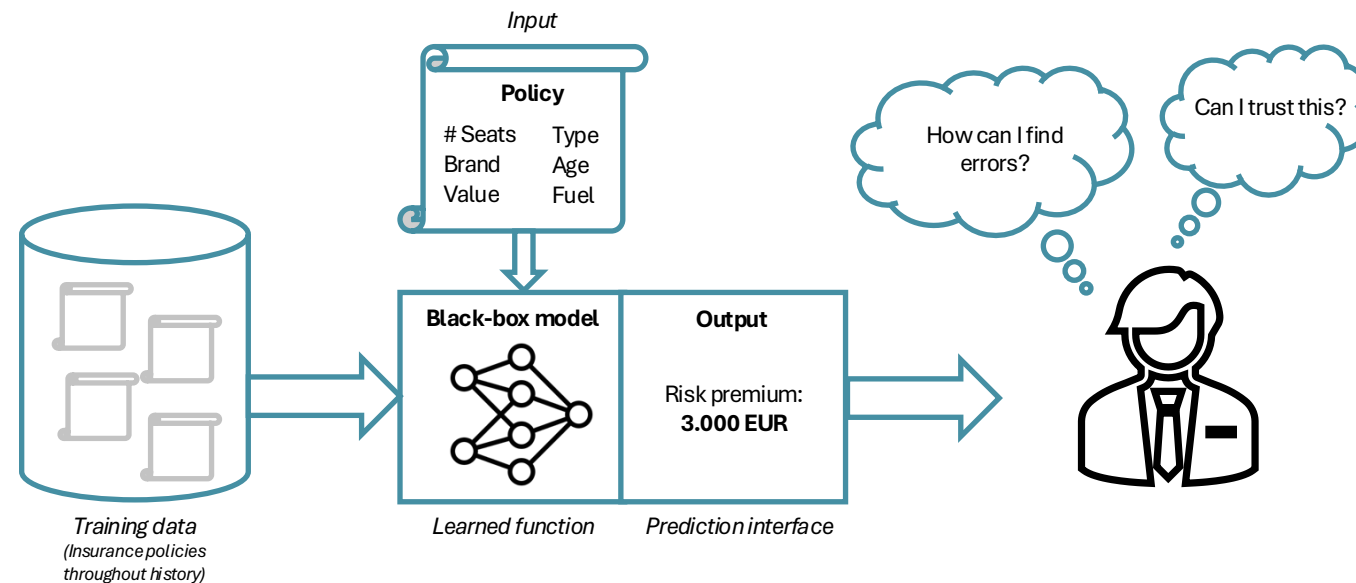
Doctoral Candidate within the
MSCA Digital Finance Action



What is XAI and why does it matter?

XAI Explainable Artificial Intelligence

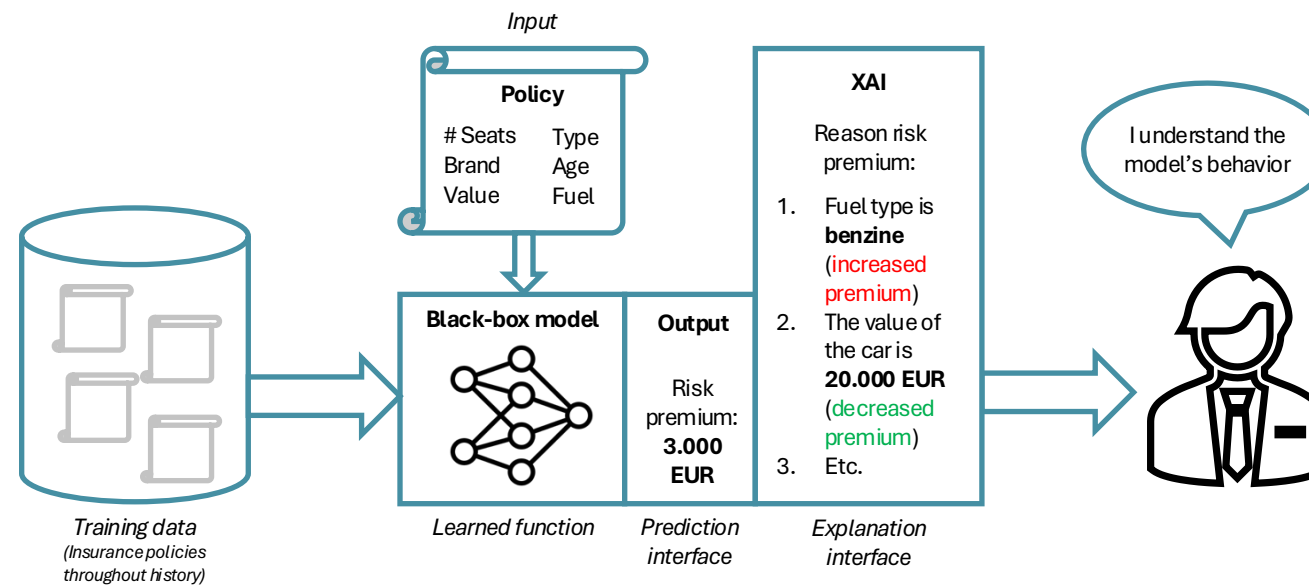
Why? XAI is critical because black-box models dominate, but understanding "why" matters more than ever



What is XAI and why does it matter?

XAI Explainable Artificial Intelligence

Why? XAI is critical because black-box models dominate, but understanding "why" matters more than ever



Background

What	A SLR was conducted within the field of temporal data, ML, and XAI
Method	PRISMA

Investigation:	Financial Time Series
	AI for Financial Time Series
	XAI for Financial Time Series
	Explainability Validation Metrics
	Limitations/Challenges of Current XAI Methods



Background

What SLR is broad, incorporating various aspects (such as statistical properties of time series)

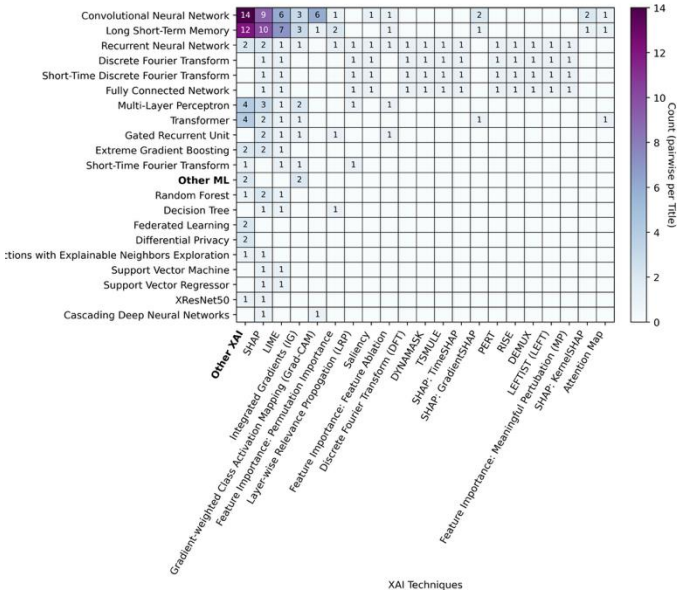
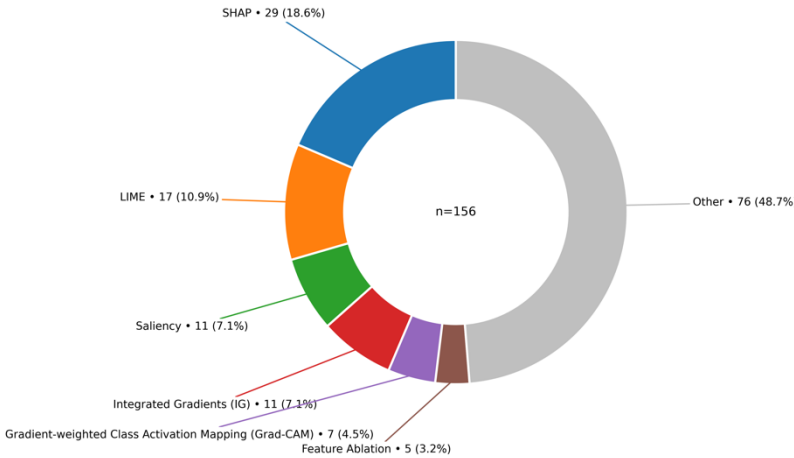
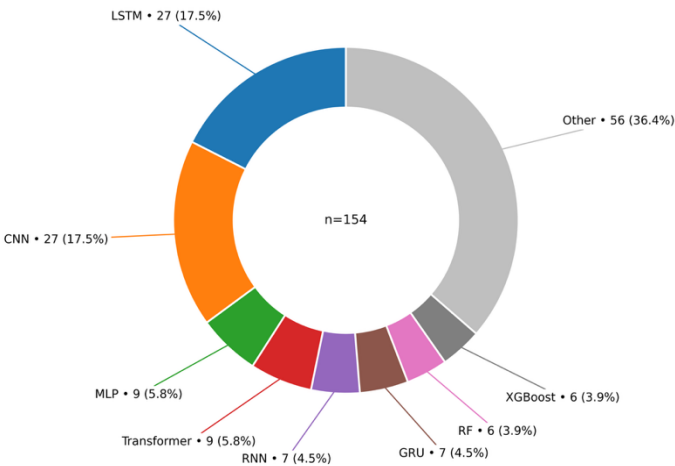
Finding Financial data are **non-linear**, **evolving**, and **dependent over time** (properties fit all to a certain degree)

	Statistical property	Explanation
●	Temporal dependencies (sequential nature)	Refers to the relationship between financial time series' time points.
●	Time-varying dependence structure	Refers to a non-constant joint distribution of variables over time.
●	Non-stationarity	Indicates that the statistical properties of the financial time series, such as mean and variance, change over time.
●	Trend, seasonality, irregularity (noise and residuals)	Represent the systematic components into which a financial time series can be decomposed.
●	Autocorrelation	Measures the linear relationship between a financial time series and its lagged values, capturing how past observations influence current ones.
●	Heteroskedasticity (volatility clustering)	Describes time-varying variance in financial time series, where periods of high volatility tend to cluster together.
●	Heavy tails (fat-tailedness)	Refers to the empirical observation that return distributions often exhibit higher kurtosis than the normal distribution.
●	Regime shifts / structural breaks	Sudden changes in the financial time series due to macroeconomic events, policy changes, or crises.

Background

What

An investigation was also conducted into (X)AI for financial time series and its limitations



(Way more explanations have been generated in the SLR to support findings made...)

Limitations of State-of-the-Art XAI Methods

Finding The review showed that XAI methods have many limitations, which can be grouped into two general groups:



Technical Gap

Current XAI methods fail to account for the statistical properties of financial time series

Human-centric Gap

Explanations are not designed around the interpretive needs of financial stakeholders and their decision contexts



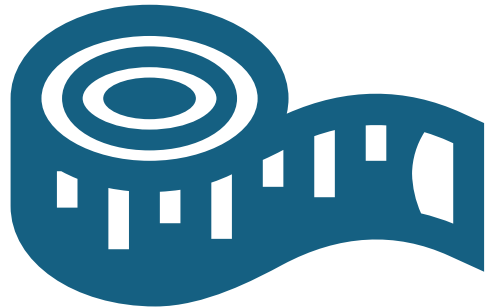
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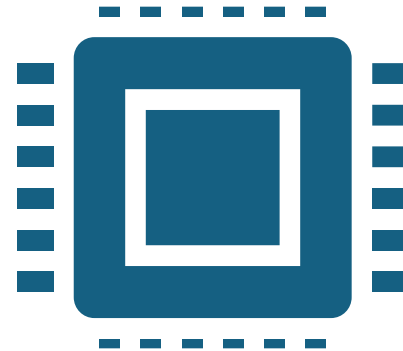
AI methods not designed for financial time series

Most techniques assume i.i.d. data and fail under temporal dependencies



Unstable and inconsistent explanations

Small changes in data or model parameters can drastically alter attributions



Computational inefficiency in real-world settings

Some XAI methods (e.g., SHAP) are too slow or resource-intensive for large financial datasets

Human-centric Gap

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Lack of domain-tailored XAI methods for finance

No existing frameworks or techniques explicitly incorporate time-aware or finance-specific properties

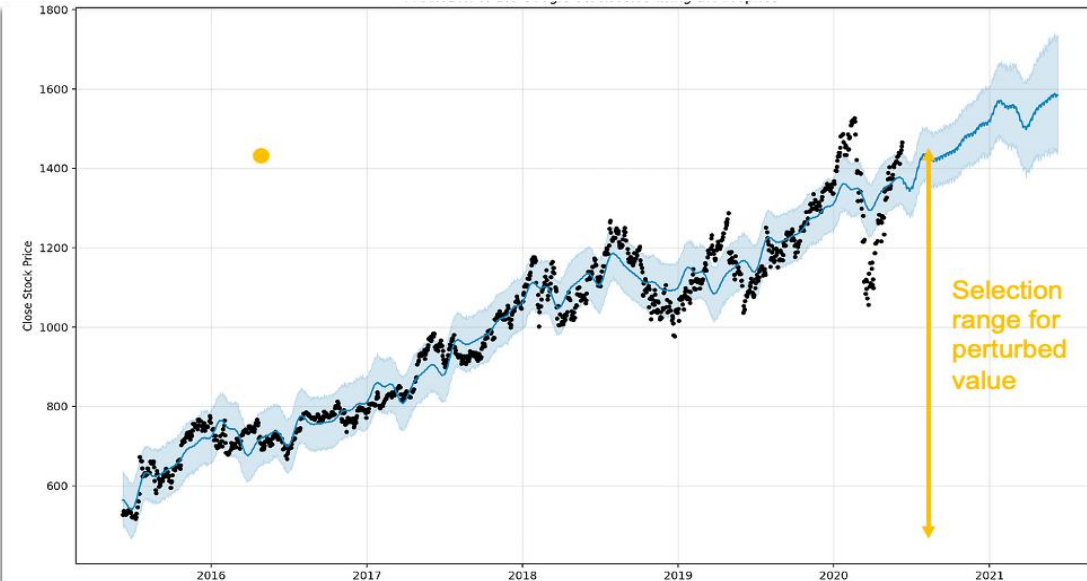
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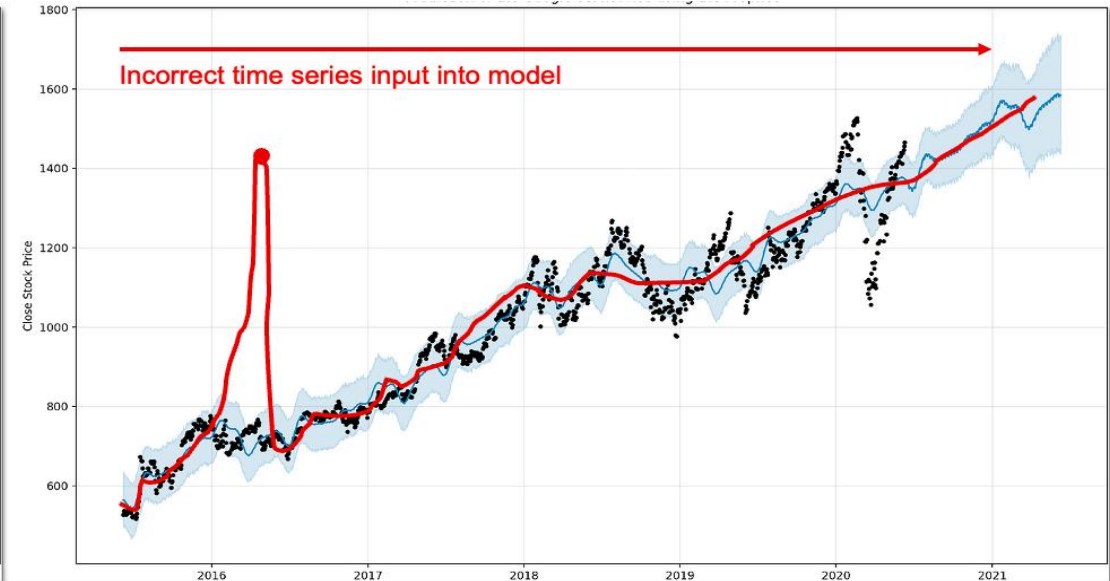
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Example: perturbation for XAI in time series (e.g., SHAP)

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Explanations not aligned with stakeholder needs

Regulators, risk managers, traders, and data scientists each require different forms of explainability



Lack of user-centered evaluation in explanations

Explanations are rarely tested for how humans perceive, trust, or use them in decisions



No mapping between explainability metrics and stakeholder priorities

Metrics such as fidelity, stability, or robustness are not linked to what each stakeholder values



Operational irrelevance of mathematically correct explanations

Even accurate explanations may be unhelpful or misleading in real-world financial workflows

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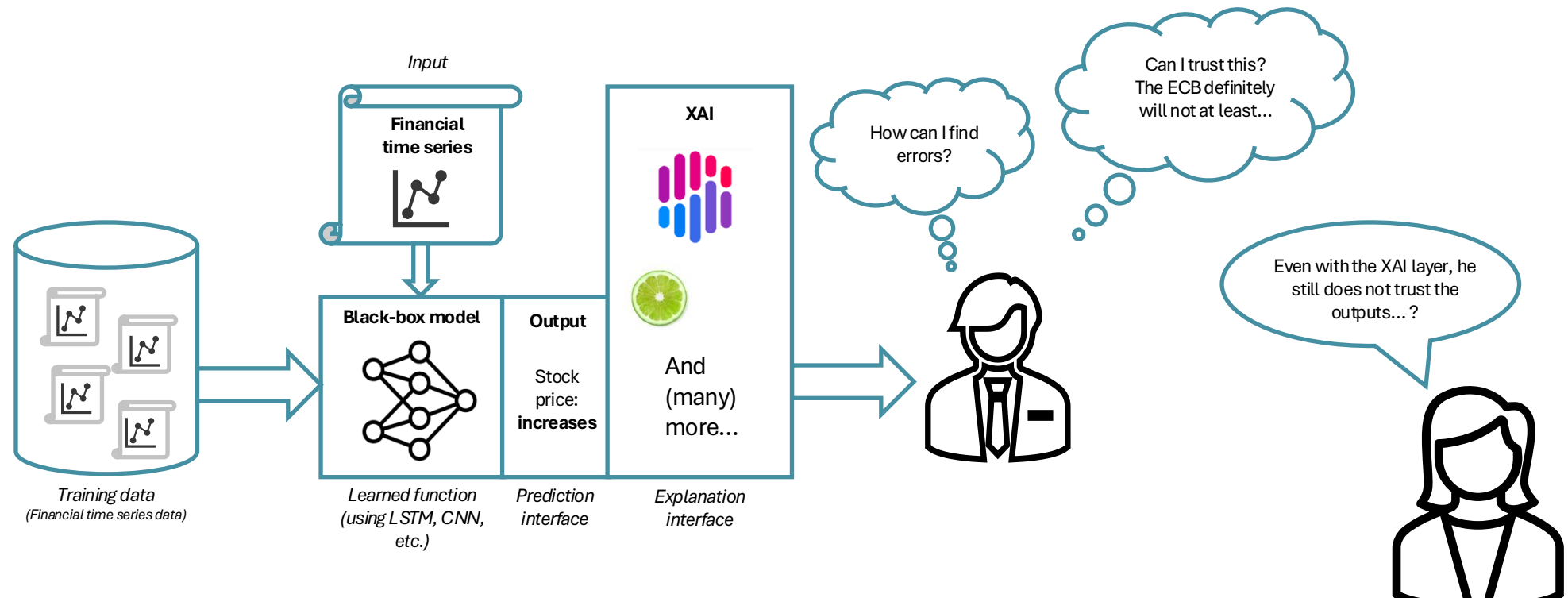


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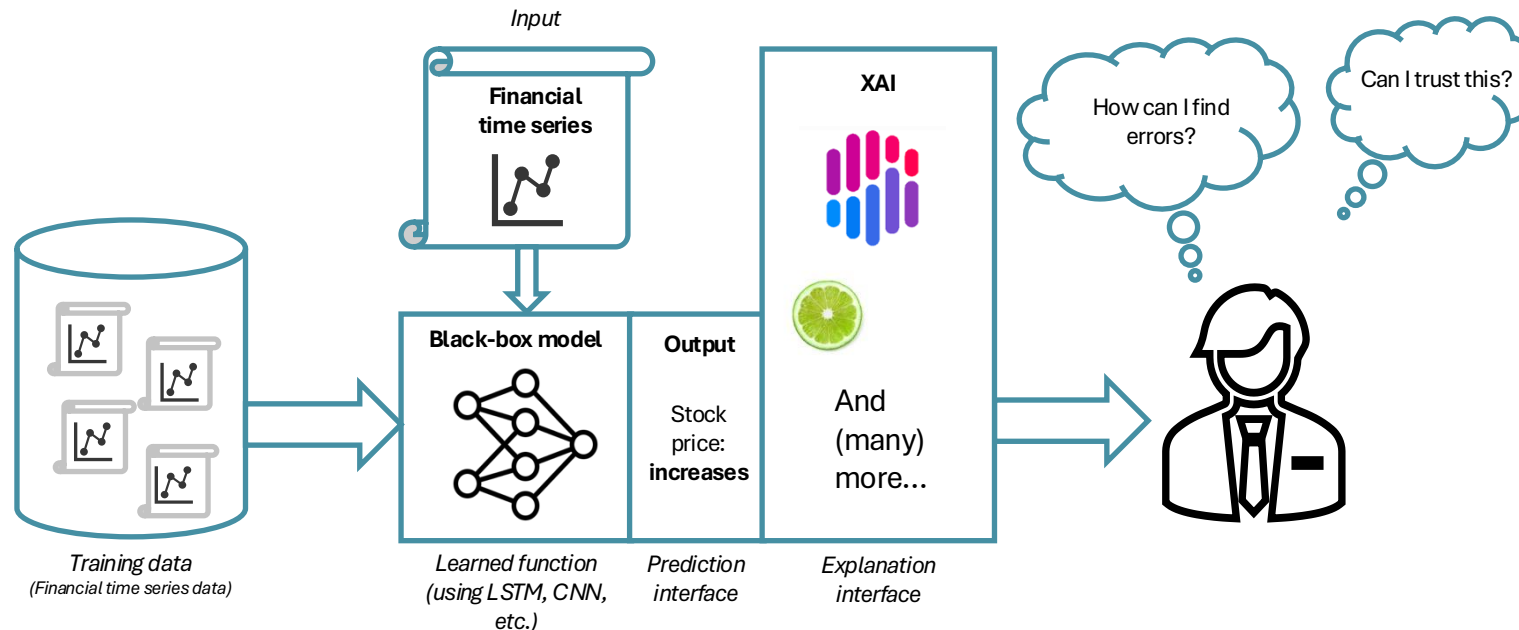


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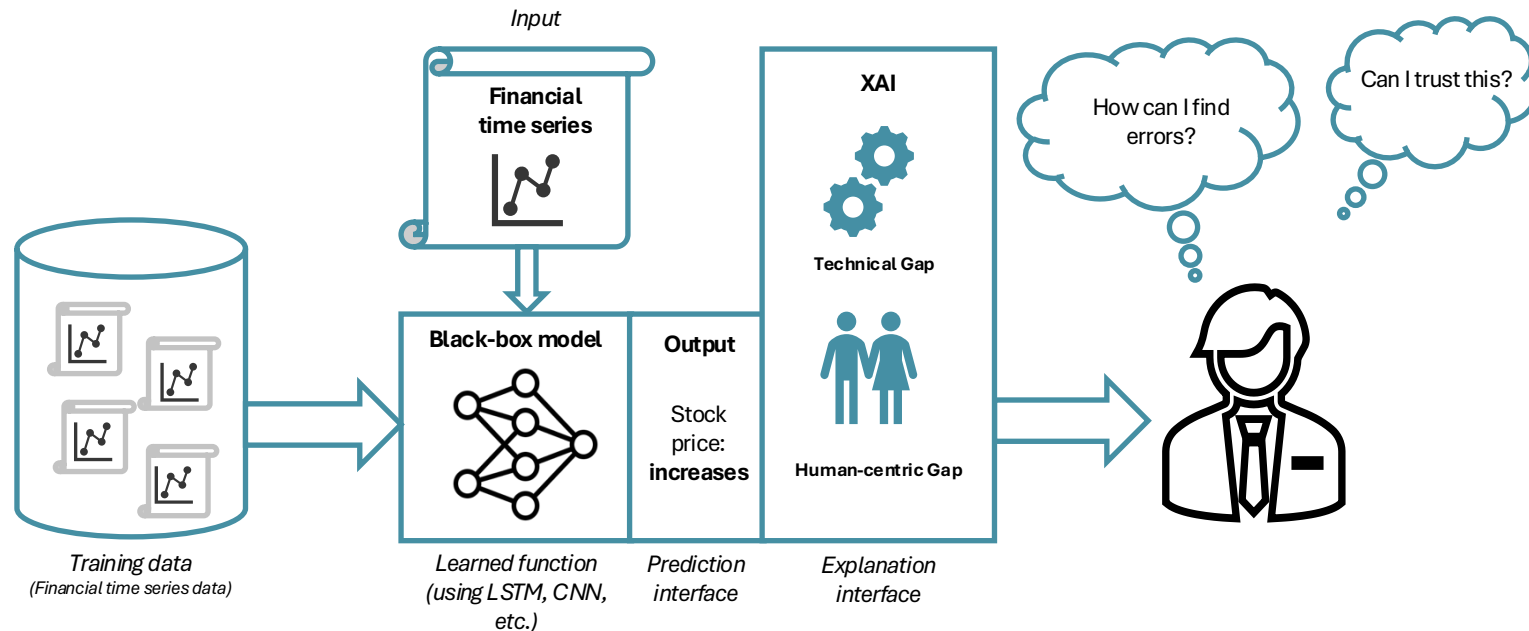
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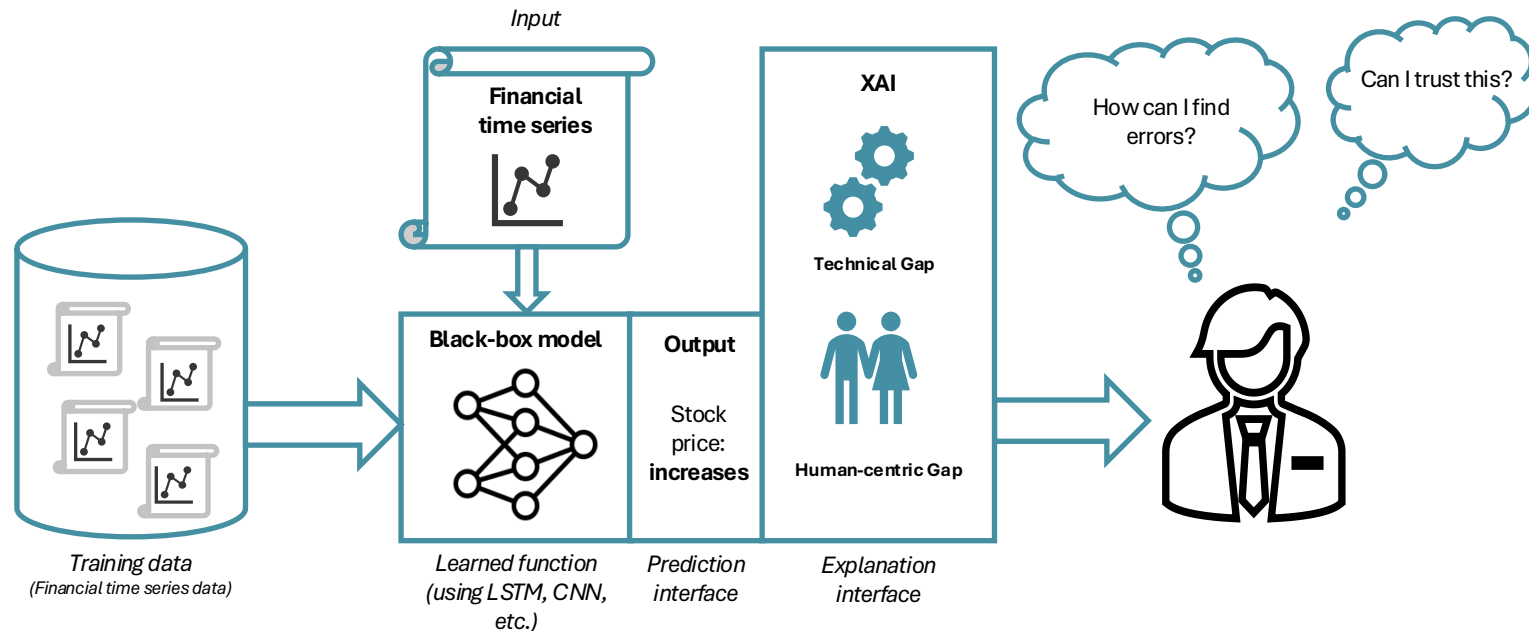
How to proceed from here?



Limitations of State-of-the-Art XAI Methods

Finding The review showed that XAI methods have many limitations, which can be grouped into two general groups

Will be discussed in a follow-up presentation...



Asking for your input!

Require We are very happy to collect your input (!) on the identified limitations in the field



Decarbonizing the S&P 500: A Sector-Level Analysis Under Tracking Error Constraints

Rebecca Di Francesco

Doctoral Candidate within the
MSCA Digital Finance Action



Context and Research Gap

1

Context: Why the Topic Matters

- **Climate change** -> societal challenge but also **systemic financial risk** (physical risk and transition risk)
- **Portfolio decarbonization** emerge as **risk hedging tool**

2

Research Gap

- Many studies explore portfolio decarbonization at the **global portfolio level** instead of adopting a **sectoral perspective**
 - **Financial performance** still important -> **Tracking Error (TE)** constraint introduced
- No existing study assesses **sector-specific decarbonization potential under tracking-error constraints** and compares the **key characteristics that drive differences across sectors**

Research Questions & Contribution

3

Research Questions

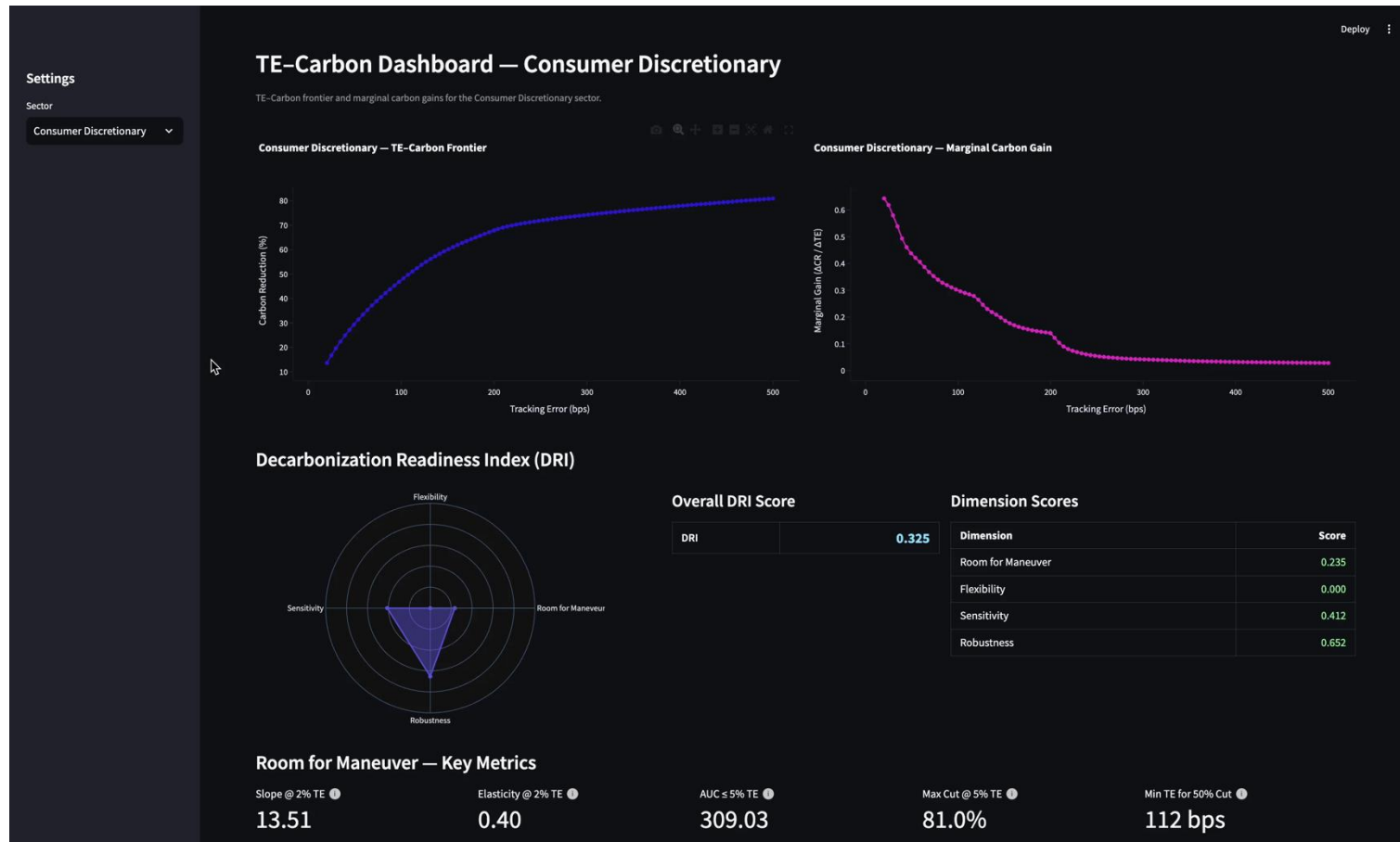
- Which **sectors** are structurally **most** and **least ready to decarbonize** under **tracking-error constraints**?
- How does **decarbonization readiness** for each sector emerge from the joint evaluation of **room for maneuver, flexibility, sensitivity, and robustness**?

4

Contribution

- Practical tool for investors (**dashboard**) to guide **portfolio design**
- **Policy makers** can also get informed about which **sectors** need more **capital** to support the **green transition**

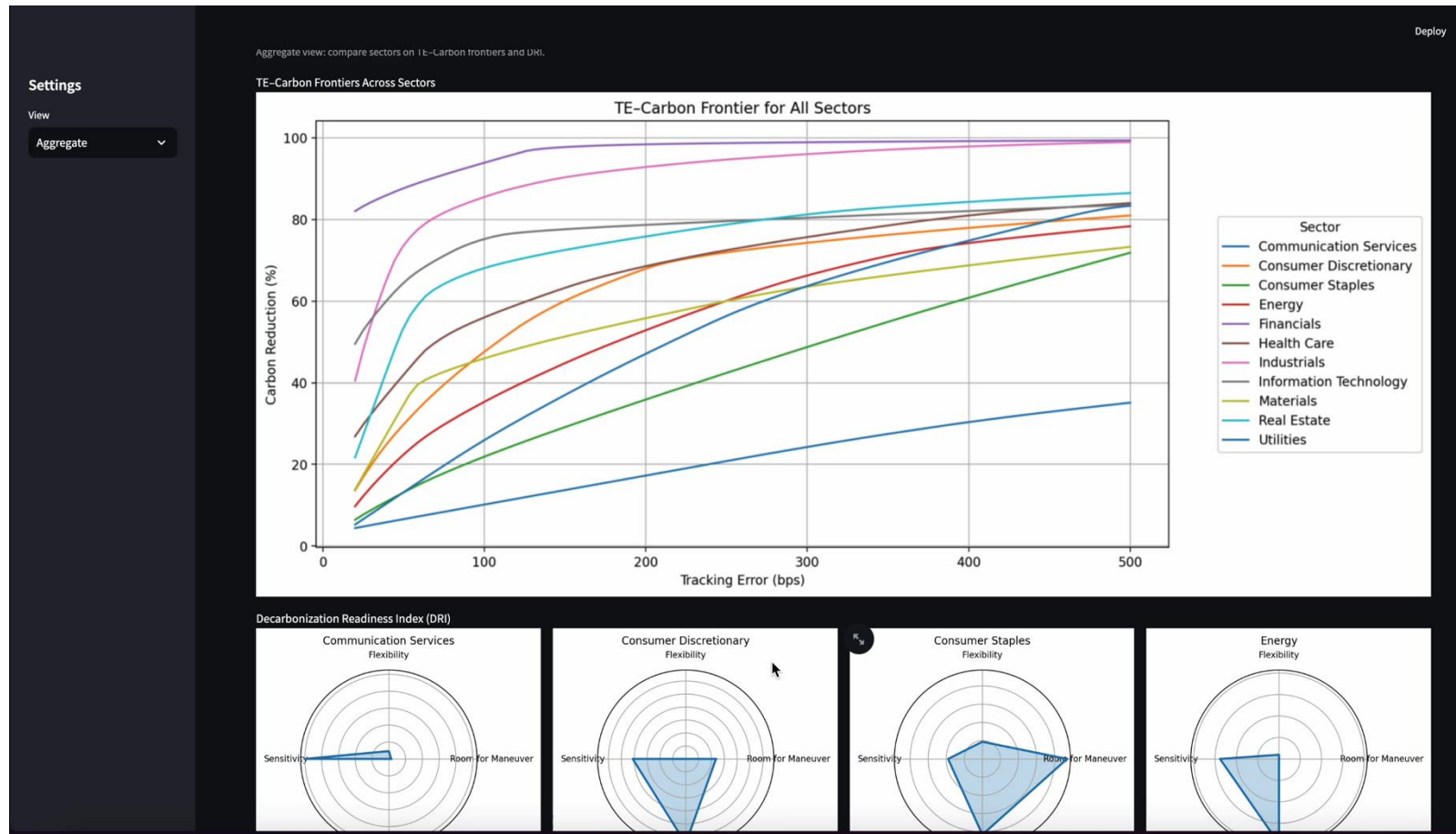
Dashboard: Sector-Level View



The dashboard will offer an individual sector view displaying:

- The **TE-Carbon Frontier**
- **All key dimension metrics and scores:** Room for Maneuver, Flexibility, Sensitivity, and Robustness
- The resulting **Decarbonization Readiness Index (DRI)**

Dashboard: Aggregate Sector View



The dashboard will also provide an **aggregate** view enabling:

- Comparison of **TE-Carbon frontiers across sectors**
- Comparison of **key dimensions** and **DRI scores across sectors**

**Feedback is
appreciated!**





DIGITAL



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